

Astra, Opus and the revolved mesher

Four revolved profiles

Prepared by Ryan | 8 October 2026

Independent LLM block layouts and deterministic CAD-to-mesh results for four synthetic revolved profiles. Features include steps, chamfers, circular fillets and two through bores with distinct inner and outer contours.

Part	Approach	Status	Hexes	Min SJ / doubled	Checks	Time	Usage*
01	GPT 6 Astra	Pass	4,500	0.6964 / 0.7006	1	2.28 m	\$1.02
01	Opus 5.5	Pass	3,584	0.6946 / 0.6996	1	4.03 m	\$0.93
01	Revolved mesher	Pass	6,180	0.6855 / 0.6860	1	0.043 s	No API
02	GPT 6 Astra	Pass	1,824	0.5774 / 0.5774	1	2.22 m	\$0.93
02	Opus 5.5	Pass	1,740	0.6428 / 0.6428	1	6.55 m	\$1.34
02	Revolved mesher	Refused	-	- / -	0	0.002 s	No API
03	GPT 6 Astra	Pass	2,400	0.5987 / 0.5997	1	1.75 m	\$1.04
03	Opus 5.5	Pass	1,216	0.6428 / 0.6452	1	4.54 m	\$1.13
03	Revolved mesher	2x failure	3,528	0.7011 / -	1	0.021 s	No API
04	GPT 6 Astra	Pass	2,880	0.5761 / 0.5126	1	2.60 m	\$1.07
04	Opus 5.5	Pass	1,728	0.6446 / 0.6456	1	5.04 m	\$1.21
04	Revolved mesher	2x failure	3,936	0.2566 / -	1	0.020 s	No API

*GPT dollars are Standard API-equivalent estimates from cumulative native tokens; Claude dollars are CLI-reported usage. Neither is a subscription invoice. Deterministic time covers compilation; model time covers the author session. Astra uses medium reasoning, Opus high.

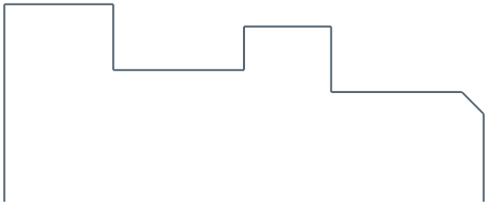
Explicitly synthetic showcase, not a population benchmark. The original CAD and native geometry checks are shared within this experiment. Automatic-route failures and refusals remain visible. No prior or other-model layouts supplied to authors; no expert grades assigned.

Multi-step shaft with chamfered tip

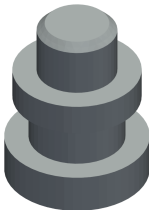
Original CAD

Full solid of revolution.
No holes or internal cavities.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality

Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

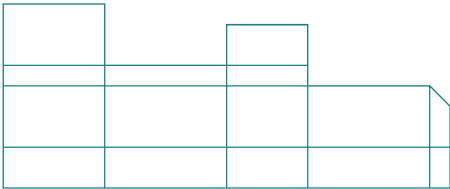
GPT 6 Astra

Geometry pass

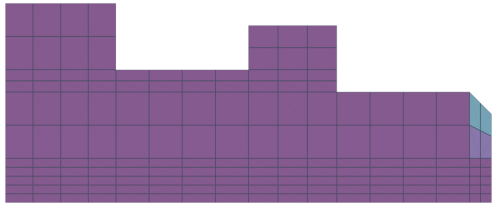
17 seed parts | 4500 hexes
SJ 0.6964 / 2x 0.7006
1 full checks | 2.28 min
Usage \$1.02

Standard API equivalent

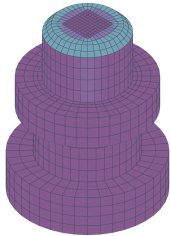
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



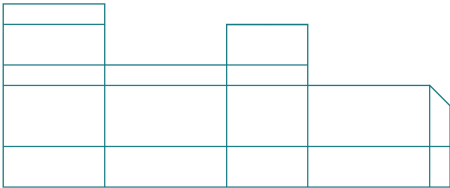
Opus 5.5

Geometry pass

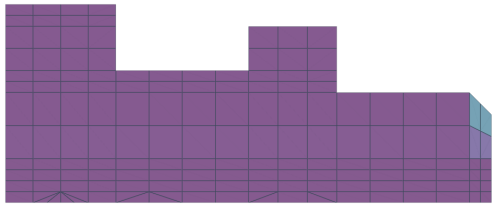
1 seed parts | 3584 hexes
SJ 0.6946 / 2x 0.6996
1 full checks | 4.03 min
Usage \$0.93

Claude CLI usage

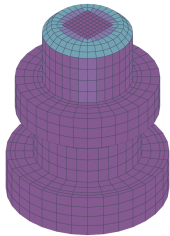
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



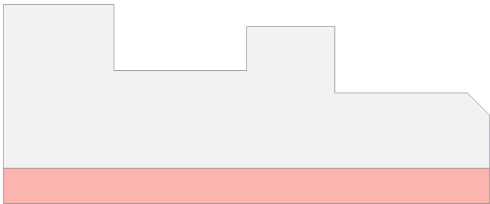
Revolved mesher

Geometry pass

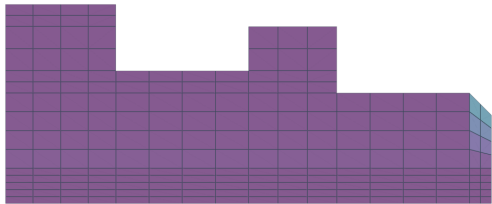
5 seed parts | 6180 hexes
SJ 0.6855 / 2x 0.6860
1 full checks | 0.043 s
Usage no API

local deterministic computation

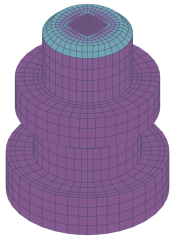
REALIZED SEED PARTS



MESH SECTION / SJ



3D MESH / SJ



SJ 0.2-1; larger is better. Seed parts may contain several blocks.

Filletted tapered body with rounded ends

Original CAD

Full solid of revolution.
No holes or internal cavities.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality

Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

Fillet detail: Astra represents the tiny fillet with 48/192 boundary quads; Opus spans it (0 attributed quads). Both pass native checks; neither has an eligible sharp edge in this case.

GPT 6 Astra

Geometry pass

27 seed parts | 1824 hexes
SJ 0.5774 / 2x 0.5774
1 full checks | 2.22 min
Usage \$0.93

Standard API equivalent

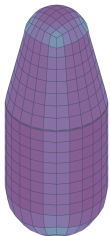
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



Opus 5.5

Geometry pass

1 seed parts | 1740 hexes
SJ 0.6428 / 2x 0.6428
1 full checks | 6.55 min
Usage \$1.34

Claude CLI usage

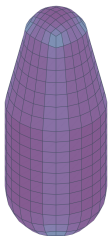
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



Revolved mesher

Refused

- seed parts | - hexes
SJ - / 2x -
0 full checks | 0.002 s
Usage no API

local deterministic computation

No layout or mesh emitted.

pole-revolve also declined: pole-revolve: shallow-junction-13.1deg (two-surface ring inheritance converges at rate $\cos^2$ of the junction angle under a 50-iteration cap, so at 13.1° it does not converge; a tangent or shallow junction needs every ring node pinned, which this construction has no mechanism for)]



SJ 0.2-1; larger is better. Seed parts may contain several blocks.

Hollow flanged sleeve with stepped bore

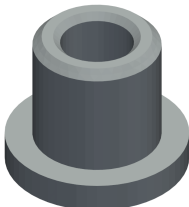
Original CAD

Through-bored solid of revolution.
Inner and outer profiles.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality

Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

Automatic mesher: nominal geometry passes, but the doubled run stops on inverted cells. Nominal views are shown; no doubled mesh or SJ is available.

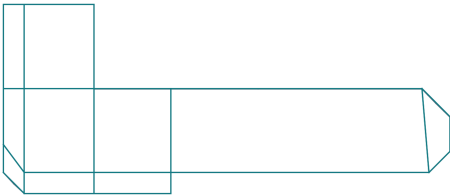
GPT 6 Astra

Geometry pass

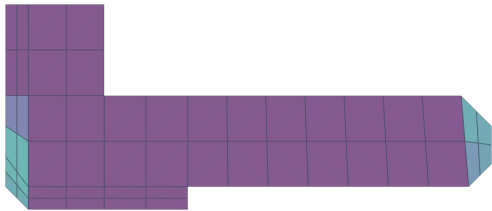
40 seed parts | 2400 hexes
SJ 0.5987 / 2x 0.5997
1 full checks | 1.75 min
Usage \$1.04

Standard API equivalent

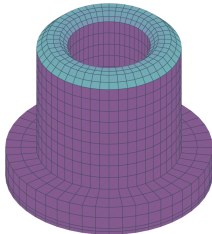
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



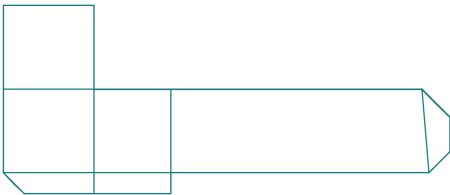
Opus 5.5

Geometry pass

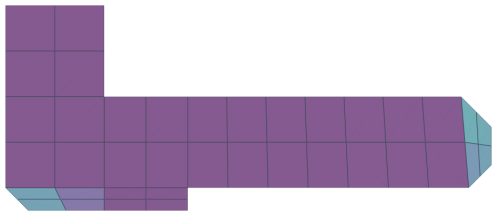
1 seed parts | 1216 hexes
SJ 0.6428 / 2x 0.6452
1 full checks | 4.54 min
Usage \$1.13

Claude CLI usage

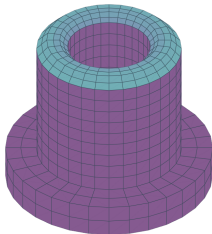
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



Revolved mesher

Refinement failure

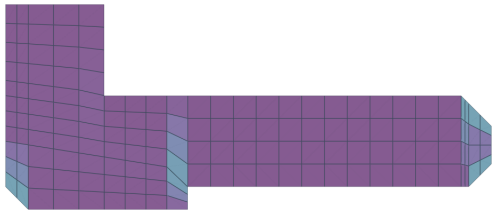
4 seed parts | 3528 hexes
SJ 0.7011 / 2x -
1 full checks | 0.021 s
Usage no API

local deterministic computation

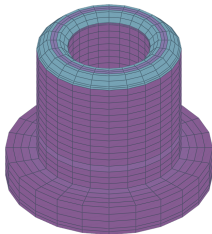
REALIZED SEED PARTS



MESH SECTION / SJ



3D MESH / SJ



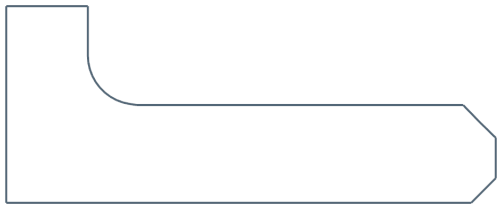
SJ 0.2-1; larger is better. Seed parts may contain several blocks.

Hollow collar with rounded shoulder

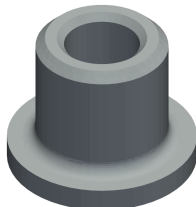
Original CAD

Through-bored solid of revolution.
Inner and outer profiles.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality

Teal partitions are author plans. Colored seed sections and SJ mesh cuts come from the actual 3D mesh. Both densities are checked.

Automatic mesher: nominal geometry passes, but the doubled run stops on inverted cells. Nominal views are shown; no doubled mesh or SJ is available.

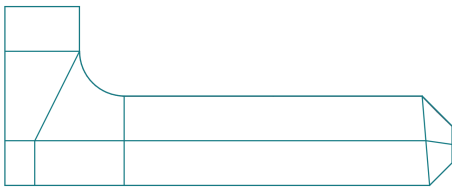
GPT 6 Astra

Geometry pass

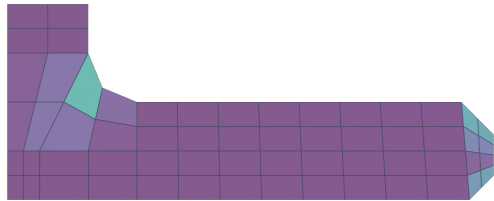
72 seed parts | 2880 hexes
SJ 0.5761 / 2x 0.5126
1 full checks | 2.60 min
Usage \$1.07

Standard API equivalent

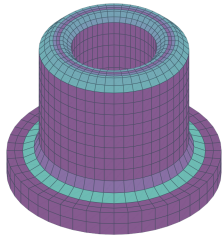
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



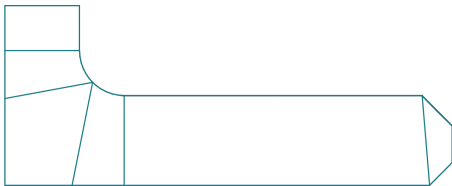
Opus 5.5

Geometry pass

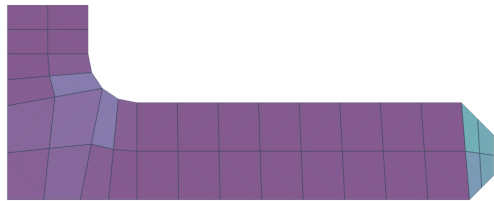
1 seed parts | 1728 hexes
SJ 0.6446 / 2x 0.6456
1 full checks | 5.04 min
Usage \$1.21

Claude CLI usage

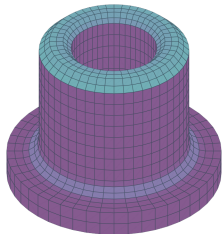
PLANNED PARTITION



MESH SECTION / SJ



3D MESH / SJ



Revolved mesher

Refinement failure

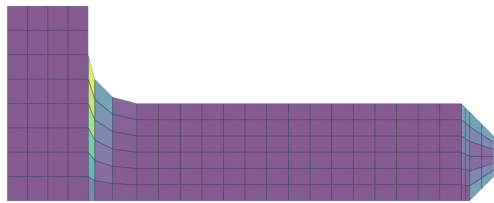
4 seed parts | 3936 hexes
SJ 0.2566 / 2x -
1 full checks | 0.020 s
Usage no API

local deterministic computation

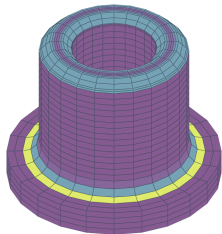
REALIZED SEED PARTS



MESH SECTION / SJ



3D MESH / SJ



SJ 0.2-1; larger is better. Seed parts may contain several blocks.